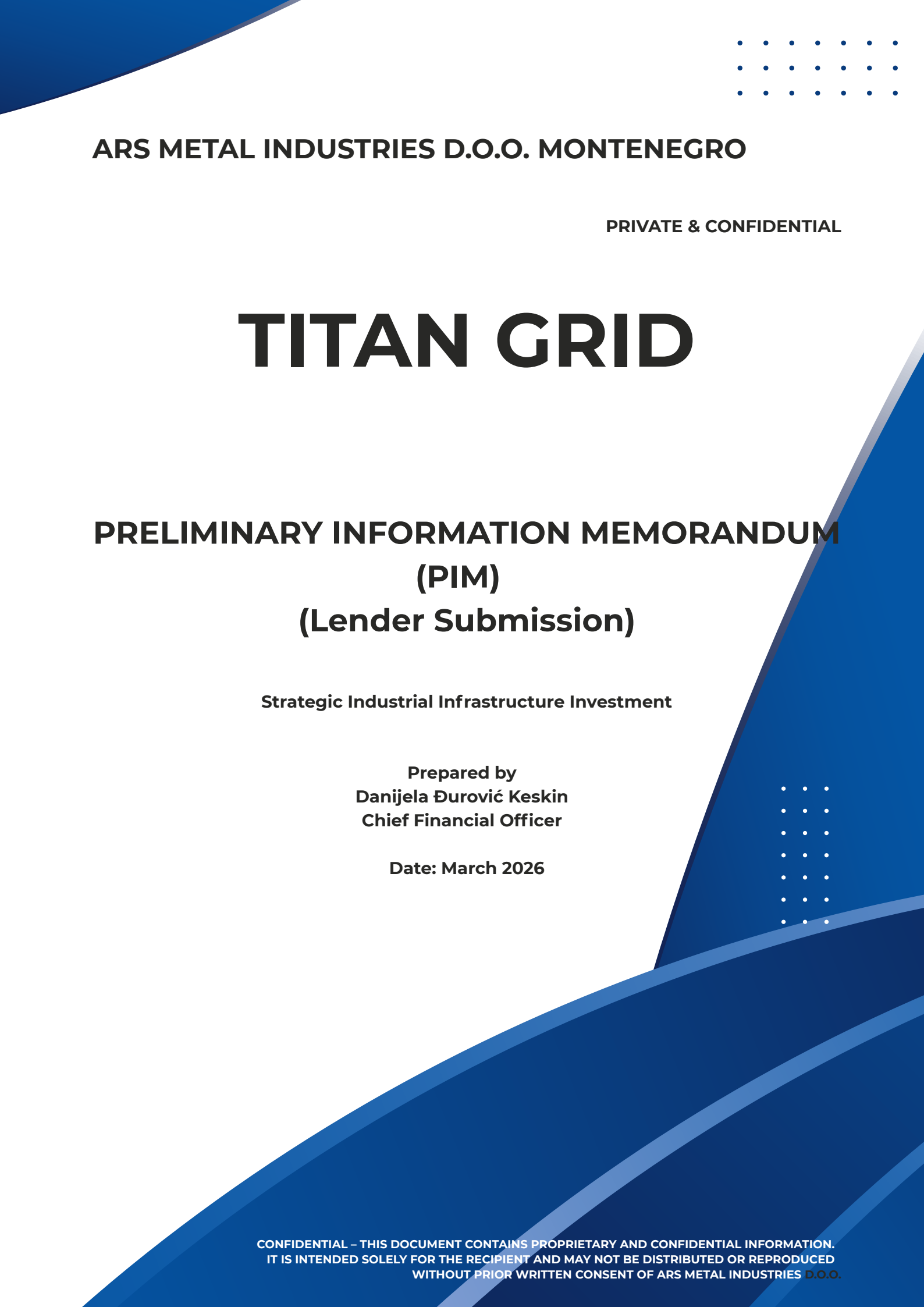




ARS METAL INDUSTRIES D.O.O. MONTENEGRO

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TITAN GRID

PRELIMINARY INFORMATION MEMORANDUM (PIM) (Lender Submission)

Strategic Industrial Infrastructure Investment

**Prepared by
Danijela Đurović Keskin
Chief Financial Officer**

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EXECUTIVE SUMMARY: THE TITAN GRID OPPORTUNITY

TITAN Grid is a purpose-built Industry 4.0 manufacturing platform located in Tuzi, Montenegro.

Structured as a fully ring-fenced special purpose vehicle (SPV) under ARS Metal Industries d.o.o., the facility is designed to produce high-precision power transformer tanks and radiator systems for Tier-1 OEMs and transmission system operators (TSOs) across the European energy corridor. *The project addresses a critical structural bottleneck in the European energy transition: the persistent shortage of high-voltage transformer components, which is currently contributing to grid expansion delays of up to 24–48 months.* Through advanced automation and localized supply chains, TITAN Grid enables a 3–7 day regional delivery model, significantly reducing logistics complexity and associated carbon exposure compared to offshore sourcing.

Strategic Highlights

Financial Resilience:

Total investment of €16.0 million, with a base case DSCR of approximately 10.7x and a downside case of approximately 2.95x

Technological Capability:

30 kW CNC laser cutting and a high degree of robotic welding support consistent quality and compliance with ISO 3834-2 standards

Lead-Time Advantage:

3–7 day delivery to Central Europe versus 90+ days from offshore suppliers, supporting potential pricing premiums and stronger customer retention

Energy Cost Stability:

Integrated 1.4 MWp solar array providing approximately 70% self-consumption, reducing exposure to grid price volatility

Regulatory Alignment:

Designed to align with CBAM requirements and EU Taxonomy principles, supporting OEM Scope 3 emissions targets

Scalable Platform:

22,742 m² site with capacity for phased expansion (TITAN 2 and TITAN 3), structured to enable partial self-financing through operating cash flow



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PAGE 2 positions TITAN Grid as an infrastructure-aligned industrial platform within the Western Balkans. It offers institutional lenders and investors exposure to a high-growth market supported by automation-driven efficiency, strong financial metrics, and alignment with European energy transition priorities. TITAN Grid represents a manufacturing platform addressing a defined supply chain constraint within a critical sector, supported by a scalable operating model and structured risk mitigation.

Formal Declaration

This Executive Summary provides a high-level overview of the TITAN Grid project. All information is supported by technical, legal, and financial documentation contained within the Preliminary Information Memorandum (PIM). Any investment decision remains subject to full due diligence and final approvals.



1. THE STRUCTURAL BOTTLENECK

European Grid Expansion Constrained by Transformer Component Supply
The European energy transition is facing a fundamental capacity constraint that cannot be resolved by capital allocation alone.

While the European Commission has earmarked approximately €584 billion for grid modernization by 2030, project execution is increasingly constrained by a shortage of power and distribution transformers. The primary limitation lies not in final transformer assembly, but in the supply of critical structural components—specifically high-precision steel tanks and specialized cooling systems.

The Global Supply Lag

European OEMs have historically relied on globalized supply chains. However, this model is under strain due to accelerating electrification demand and structural supply limitations.

Extended Lead Times:

Delivery timelines for standard transformers have increased from approximately 6–8 months to 24–48 months.

Supply Chain Exposure:

Dependence on suppliers in Asia and Turkey involves 60–90 day maritime transport, increasing exposure to geopolitical disruptions, logistical bottlenecks, and freight cost volatility.

Production Backlogs:

Tier-1 manufacturers (e.g., Siemens Energy, Hitachi Energy, GE Vernova) face significant order backlogs, where delays in a single critical component can stall high-value grid infrastructure projects.

The Technical Constraint

Transformer tanks are high-specification engineered structures requiring precision manufacturing under strict quality standards. The bottleneck is further intensified by limited regional advanced manufacturing capacity:

Skilled Labor Shortage:

Traditional production relies on highly specialized manual welders, a workforce segment in decline across Europe.

Quality Variability:

Manual processes can result in rework rates of approximately 8–12%, contributing to delays and cost inefficiencies.

Regulatory Pressure:

The EU's Carbon Border Adjustment Mechanism (CBAM) is increasing the cost and complexity of sourcing carbon-intensive steel components from non-EU markets.

The TITAN Grid Response

TITAN Grid is positioned to address this structural constraint through a high-automation, near-shore production model in Tuzi, Montenegro:

Near-Shoring Advantage: Replaces extended maritime supply chains with 3–7 day road delivery to Central European markets.

Automated Production:

High levels of robotic welding reduce dependence on scarce manual labor and improve production consistency.

Lower-Carbon Manufacturing:

Integration of a 1.4 MWp on-site solar system supports reduced carbon intensity and alignment with CBAM requirements.

Strategic Perspective

TITAN Grid represents an industrial solution to a clearly defined supply chain constraint within Europe's energy transition. By reducing lead times and improving supply reliability, the project supports more efficient deployment of grid infrastructure in a capacity-constrained market.

2. MARKET ANALYSIS & COMPETITIVE LANDSCAPE

Capturing the €584 Billion Grid Modernization Opportunity

The investment case for TITAN Grid is anchored in a structural, multi-decade upgrade of Europe's power infrastructure. Under the EU Action Plan for Grids, the European Commission has identified an estimated €584 billion investment requirement by 2030. This investment is not discretionary; it underpins the execution of REPowerEU and Fit-for-55 objectives.

2.1 ADDRESSABLE MARKET: STRUCTURAL DEMAND EXPANSION

The transformer market is transitioning from a replacement-driven cycle to a structurally driven expansion phase.

- **Grid Modernization:** More than 50% of Europe's low-voltage network is expected to exceed 40 years of service by 2030, requiring accelerated replacement of transformers and associated components
- **Renewable Integration:** Each 1 GW of new wind or solar capacity requires significant transformer infrastructure. High-capacity units (above 100 MVA) are projected to grow at a CAGR of approximately 8.78% through 2031
- **Electrification of Transport:** The TEN-T revision targets 55.6 GW of public charging capacity by 2030, implying the need for over 150,000 additional distribution transformers to support high-capacity charging infrastructure

2.2 COMPETITIVE LANDSCAPE & NEAR-SHORING DYNAMICS

The supply landscape is characterized by constraints at both ends of the market:

- **Global OEMs (e.g., Siemens Energy, Hitachi Energy, ABB, Schneider Electric):** These players face record order backlogs. The primary constraint lies in component supply rather than demand. TITAN Grid is positioned as a complementary supplier within this ecosystem
- **Regional Fabricators:** Smaller manufacturers, primarily in Turkey and Eastern Europe, often rely on manual processes and lack the advanced manufacturing capabilities required for high-specification components
- **Competitive Positioning**
- **Lead-Time Advantage:** Offshore sourcing can involve extended delivery timelines for complex components, whereas TITAN Grid enables delivery within 3–7 days to key European markets
- **Reliability & Responsiveness:** Reduced lead times and localized production support improved project execution for OEMs and grid operators

2.3 REGULATORY CONTEXT: CBAM & ESG ALIGNMENT

The Carbon Border Adjustment Mechanism (CBAM), entering its implementation phase, is reshaping the competitive landscape for industrial inputs:

- **Carbon Cost Exposure:** Imports from higher-emission jurisdictions are increasingly subject to carbon-related cost adjustments linked to EU ETS benchmarks
- **Positioning Advantage:** Lower-carbon production, supported by on-site solar generation and regional sourcing, reduces exposure to CBAM-related costs and supports alignment with OEM Scope 3 emissions targets

Market Conclusion

The transformer component market is increasingly driven by structural demand linked to energy transition policies rather than traditional industrial cycles. Supply constraints and extended lead times are contributing to a supply-demand imbalance.

Within this context, TITAN Grid is positioned to serve an existing and undersupplied market by providing localized, high-specification manufacturing capacity aligned with evolving regulatory and operational requirements.

3. TECHNICAL SPECIFICATIONS & OPERATIONAL EXCELLENCE

Industry 4.0 Platform: Driving High Operational Efficiency

The core of the **TITAN Grid** investment case is the transition from labor-intensive manufacturing to a highly automated, digitally integrated production system.

By deploying 30 kW fiber laser cutting and robotic tandem welding, the project addresses key physical and operational bottlenecks associated with traditional fabrication models.

The targeted Overall Equipment Effectiveness (OEE) of >85% is supported by integrated automation, process optimization, and real-time monitoring systems.

3.1 THE 30 kW CNC FIBER LASER: MATERIAL PROCESSING CAPABILITY

The production process begins with high-power CNC fiber laser technology, enabling efficient and precise material processing:

- **High Throughput:** Capable of cutting 20 mm carbon steel at speeds exceeding 2.5 m/min, significantly outperforming conventional 6–10 kW systems commonly used by regional manufacturers
- **Precision Output:** Achieves tolerances of approximately ± 0.05 mm, reducing the need for secondary processing and lowering work-in-progress (WIP) time
- **Cut Quality:** Nitrogen-assisted cutting produces oxide-free edges suitable for high-specification coating requirements and OEM standards

3.2 ROBOTIC WELDING SYSTEMS: CONSISTENCY AND SCALE

Robotic welding systems provide consistency, repeatability, and improved cycle efficiency:

- **Automated Welding:** Multi-axis robotic systems enable high-quality welds with reduced reliance on manual labor
- **Cycle Time Efficiency:** Automated processes significantly reduce production cycle times compared to manual welding methods
- **Process Control:** Integrated monitoring systems support consistent weld quality and reduce rework rates

3.3 DIGITAL INTEGRATION & OEE PERFORMANCE

Operational efficiency is supported through the integration of digital systems and process control:

- **Availability:** Predictive maintenance systems support high equipment uptime by identifying potential issues before failure
- **Performance:** Optimized facility layout and workflow design reduce internal handling time and improve throughput
- **Quality Control:** Automated inspection and testing processes ensure consistent product quality and compliance with technical standards

Technical Conclusion

TITAN Grid represents a transition toward automated, precision-driven manufacturing within the regional industrial base. By reducing reliance on manual processes and integrating digital monitoring systems, the platform enhances production consistency, efficiency, and reliability.

This approach supports the requirements of Tier-1 OEMs for high-quality, repeatable manufacturing within a shorter and more predictable delivery framework.

4. FINANCIAL PERFORMANCE & DEBT RESILIENCE

Strong Coverage Metrics and Infrastructure-Aligned Returns

The financial structure of TITAN Grid is designed to combine operational efficiency with a conservative leverage profile. The project is projected to achieve a base case DSCR of approximately 10.7x, providing substantial headroom relative to typical lender requirements and supporting a projected equity payback period of approximately 4.1 years.

4.1 REVENUE MODEL: STABILITY AND MARGIN STRUCTURE

The revenue model incorporates multiple stabilizing factors:

- **Contracted Base Load:** A portion of production capacity is expected to be supported by framework agreements with Tier-1 OEMs, contributing to revenue visibility and baseline cash flow stability
- **Short Lead-Time Advantage:** The 3–7 day delivery capability supports responsiveness to time-sensitive orders, enhancing revenue opportunities in constrained supply conditions
- **In-House Production Capabilities:** Internal manufacturing of key components supports margin retention and reduces reliance on third-party suppliers

4.2 DEBT RESILIENCE & STRESS TESTING

The financial model has been assessed under multiple downside scenarios to evaluate resilience:

Scenario Metric	Outcome	Implication
Base Case DSCR	~10.7x	Clear safety margin beyond expected standards
Downside Case DSCR	~2.95x	Coverage remains healthy even under stress
Break-Even Utilization Capacity	<40%	Highlights the flexible use of capacity

- **Raw Material Pass-Through:** Pricing structures are designed to transfer raw material cost fluctuations to customers, reducing exposure to commodity price volatility
- **Energy Cost Stability:** On-site solar generation offsets a significant portion of energy consumption, supporting cost predictability over the long term

4.3 CAPITAL STRUCTURE & PAYBACK PROFILE

- The project is structured to balance return optimization with prudent leverage:
- **Total Investment:** €16.0 million
- **Senior Debt Range:** €6.5 million – €8.0 million (approximately 50% leverage)

- **Stabilized Revenue:** €35–37 million annually
- **Equity Payback Period:** Approximately 4.1 years from commercial operations

Investment in durable, long-life industrial assets supports sustained cash flow generation beyond the initial debt amortization period.

Financial Conclusion

TITAN Grid presents a financial profile characterized by strong cash flow generation, high debt service capacity, and resilience under downside scenarios. The combination of operational efficiency, cost control mechanisms, and conservative leverage supports a stable credit profile and attractive returns within an infrastructure-aligned industrial framework.

5. STRATEGIC IMPACT & ESG COMPLIANCE

Decarbonization and Long-Term Sustainability Alignment

TITAN Grid is designed as an industrial platform aligned with the requirements of the energy transition. By integrating renewable energy, resource efficiency, and regulatory compliance, the project supports lower-carbon manufacturing and contributes to the decarbonization objectives of its customers.

The project is aligned with the EU Green Deal and the REPowerEU framework, supporting the development of a more resilient and lower-emission energy infrastructure.

1 SOLAR ENERGY INTEGRATION: COST STABILITY AND EMISSIONS REDUCTION

- Sustainability considerations are embedded within the facility design through the integration of on-site renewable energy:
- **Energy Coverage:** A 1.4 MWp solar installation is expected to supply approximately 70% of the facility's energy consumption, reducing reliance on external energy markets.
- **Emissions Reduction:** The solar system is estimated to offset approximately 1,200 tonnes of CO₂ annually, contributing to a lower operational carbon footprint.
- **Operational Optimization:** Production scheduling can be aligned with peak solar generation periods to improve energy efficiency.

2 REGULATORY CONTEXT: CBAM & VALUE CHAIN IMPLICATIONS

The introduction of the Carbon Border Adjustment Mechanism (CBAM) is reshaping sourcing decisions within European industrial supply chains:

- **Carbon Cost Exposure:** Imports from higher-emission jurisdictions are subject to increasing carbon-related cost adjustments.
- **Lower-Carbon Production:** The use of lower-carbon inputs and renewable energy reduces exposure to CBAM-related costs.
- **Customer Alignment:** Reduced product-level carbon intensity supports OEMs in managing Scope 3 emissions within their supply chains.
- **Water Management:** The facility incorporates closed-loop water treatment systems designed to minimize discharge and support responsible resource use.

3 EU TAXONOMY ALIGNMENT & DEVELOPMENT IMPACT

TITAN Grid is designed to align with key principles of the EU Taxonomy, including “Do No Significant Harm” (DNSH), supporting eligibility for sustainable finance frameworks:

- **Land Use & Biodiversity:** A portion of the site is allocated to managed green areas, supporting environmental integration and local impact mitigation.
- **Resource Efficiency:** Advanced manufacturing processes support high material utilization and reduced waste generation.
- **Employment Impact:** The project contributes to the creation of skilled industrial roles, supporting regional economic development.

Strategic Conclusion

TITAN Grid represents an industrial asset aligned with evolving environmental standards and regulatory frameworks. By integrating energy efficiency, emissions reduction, and responsible resource management, the project supports long-term operational sustainability and compliance within the European energy transition context.

6. INVESTMENT CONVICTION & NEXT STEPS

Path to Financial Close

TITAN Grid is an execution-ready industrial project supported by secured land ownership, an integrated technical design, and engagement with Tier-1 OEMs. The project has progressed through key development stages and is positioned to move toward financial close.

Institutional partners are invited to participate in the final €16.0 million capital structure supporting the development of a high-specification transformer component manufacturing platform in Southeast Europe.

6.1 CAPITAL STRUCTURE SUMMARY

The financing framework is designed to balance lender protection with sponsor commitment, maintaining a conservative leverage profile:

Funding Component	Amount (Est.)	Source / Type
Senior Debt	€6.5M – €8.0M	IFI / Commercial Lenders (EIB/EBRD aligned)
Sponsor Equity	€8.0M – €9.5M	Cash + Land Contribution (Plot 413/3)
Total Investment	€16.0 million	Fully allocated

- **Asset Backing:** CAPEX investment is supported by land and core industrial equipment, providing collateral for senior lenders
- **Grant Potential:** Applications for WBIF / IPA III grants are under consideration; these are not required for base-case project viability

6.2 IMPLEMENTATION ROADMAP

Following financial close, the project is structured around a 12-month path to commercial operations:

- **Months 1–3:**
Site preparation and foundation works (Tuzi Industrial Zone)
- **Months 4–7:**
Construction of production facility and installation of solar infrastructure
- **Months 8–10:**
Installation of production equipment, including CNC laser systems and robotic welding cells
- **Month 11:**
Systems integration, testing, and commissioning
- **Month 12:**
Commercial Operations Date (COD) and initial customer deliveries

6.3 INVESTMENT RATIONALE

TITAN Grid offers a combination of operational, financial, and strategic characteristics relevant to institutional investors:

- **Strong Credit Profile:** *Projected DSCR and ring-fenced structure support lender protection*
- **Market Positioning:** *Exposure to structural demand linked to European grid modernization*
- **Sustainability Alignment:** *Integration of lower-carbon production and alignment with evolving regulatory frameworks*

Investment Conclusion

***TITAN Grid** is positioned to address a defined supply constraint within the European energy infrastructure value chain. The project combines operational capability, financial resilience, and alignment with long-term energy transition objectives.*

Participation in the financing supports the development of regional manufacturing capacity within a strategically important sector.

FINAL STATEMENT & CLOSURE

TITAN Grid represents a strategically positioned industrial platform addressing a clearly defined constraint within Europe's energy transition.

By combining advanced manufacturing capabilities, strong financial fundamentals, and alignment with EU policy frameworks, the project is structured to deliver both operational reliability and long-term value.

The investment case is supported by validated market demand, a resilient financial model, and a scalable infrastructure platform designed to meet evolving industry requirements. TITAN Grid is positioned to contribute to the strengthening of regional supply chains while supporting broader decarbonization objectives.

We welcome the opportunity to engage with institutional partners and lenders to progress toward financial close and subsequent project execution.

FOR AND ON BEHALF OF
ARS METAL INDUSTRIES d.o.o.
Danijela Đurović Keskin
Chief Financial Officer
Date: March 2026

